

✓ YOLO Model Training and Evaluation

✓ Setup

```
from ultralytics import YOLO
import cv2
import numpy as np
from mss import mss
```

```
base_model = "yolo11s.pt"
dataset_path="mtga_data/train_data/train_data_7/dataset.yaml"
tuned_model = "yolo11s_tuned_7.pt"
```

✓ Initial Training - 7 Annotations

I started by annotating just 7 screenshots. I did not include a validation set because the goal wasn't to have a final accurate version. It was just to train the model enough that it could be used to automate the annotation process for more of the dataset using label studio's API with a prediction endpoint that hits my tuned model for first pass labeling.

[+ Code](#)[+ Text](#)

```
model = YOLO(base_model)
```

```
# Train on annotated dataset
```

```
model.train(
    data=dataset_path,
    epochs=200,
    augment=True,
    # I initially tried with mosaic=True. It seemed like a great augmentation technique
    # but I ended up removing it because I don't want the model detecting mostly obscured cards
    mosaic=False,
    hsv_h=0.015,
    hsv_s=0.7,
    hsv_v=0.2,
    scale=0.5,
    flipplr=0.25
)
```

```
# Save the improved model
```

```
model.save(tuned_model)
```

🔗 Ultralytics 8.3.47 Python-3.11.9 torch-2.5.1+cpu CPU (13th Gen Intel Core(TM) i7-13700K)
engine\trainer: task=detect, mode=train, model=yolo11s.pt, data=mtga_data/train_data/train_data_7/dataset.yaml, epochs=200, time=None
Overriding model.yaml nc=80 with nc=4

		from	n	params	module	arguments
0		-1	1	928	ultralytics.nn.modules.conv.Conv	[3, 32, 3, 2]
1		-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
2		-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
3		-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
4		-1	1	103360	ultralytics.nn.modules.block.C3k2	[128, 256, 1, False, 0.25]
5		-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
6		-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]
7		-1	1	1180672	ultralytics.nn.modules.conv.Conv	[256, 512, 3, 2]
8		-1	1	1380352	ultralytics.nn.modules.block.C3k2	[512, 512, 1, True]
9		-1	1	656896	ultralytics.nn.modules.block.SPPF	[512, 512, 5]
10		-1	1	990976	ultralytics.nn.modules.block.C2PSA	[512, 512, 1]
11		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	0	ultralytics.nn.modules.conv.Concat	[1]
13		-1	1	443776	ultralytics.nn.modules.block.C3k2	[768, 256, 1, False]
14		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
15	[-1, 4]	1	0	0	ultralytics.nn.modules.conv.Concat	[1]
16		-1	1	127680	ultralytics.nn.modules.block.C3k2	[512, 128, 1, False]
17		-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
18	[-1, 13]	1	0	0	ultralytics.nn.modules.conv.Concat	[1]
19		-1	1	345472	ultralytics.nn.modules.block.C3k2	[384, 256, 1, False]
20		-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
21	[-1, 10]	1	0	0	ultralytics.nn.modules.conv.Concat	[1]
22		-1	1	1511424	ultralytics.nn.modules.block.C3k2	[768, 512, 1, True]
23	[16, 19, 22]	1	820956	ultralytics.nn.modules.head.Detect	[4, [128, 256, 512]]	

YOLO11s summary: 319 layers, 9,429,340 parameters, 9,429,324 gradients, 21.6 GFLOPs

```

Transferred 493/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
train: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_7\labels.cache... 7 images, 0 backgrounds, 0 corrupt:
val: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_7\labels.cache... 7 images, 0 backgrounds, 0 corrupt: 10

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'momentum' au
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to runs\detect\train
Starting training for 200 epochs...

```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/200	0G	1.125	3.274	1.133	93	640: 100% ██████████ 1/1 [00:01<00:00, 1.93s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 1/1 [00:00<00:00, 1.37it/
2/200	0G	1.172	3.326	1.141	93	640: 100% ██████████ 1/1 [00:01<00:00, 1.79s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 1/1 [00:00<00:00, 1.40it/
3/200	0G	1.016	3.068	1.152	94	640: 100% ██████████ 1/1 [00:01<00:00, 1.73s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 1/1 [00:00<00:00, 1.38it/

I didn't expect this to be sufficient for accurate classification but with some data augmentation it did surprisingly well. It is very accurate in finding most cards though it struggles with the "sick" class, sometimes failing to detect them entirely.

This is likely because sick cards are the rarest to encounter in the dataset and with only 7 fully annotated images we probably don't have enough.

Train 7 Screenshot

Here you can see that the model is failing to detect one of the sick cards on the opponent's side. However it is otherwise quite accurate considering how limited the dataset was. 

Training Pass #2 - 50 Annotations

With the automatic labeling, it took less time to correct the labels another 43 screenshots than it took to fully label the first 7. Now I will reserve 5 images for the validation set to make sure the model isn't overfitting.

```

dataset_path="mtga_data/train_data/train_data_50/dataset.yaml"
tuned_model = "yolo11s_tuned_50.pt"

```

```
model = YOLO(base_model)
```

```
# Train on annotated dataset
```

```

model.train(
    data=dataset_path,
    epochs=200,
    augment=True,
    mosaic=False,
    hsv_h=0.015,
    hsv_s=0.7,
    hsv_v=0.2,
    scale=0.5,
    flip_lr=0.25
)

```

```
# Save the improved model
```

```
model.save(tuned_model)
```

```

→ Ultralytics 8.3.47 Python-3.11.9 torch-2.5.1+cpu CPU (13th Gen Intel Core(TM) i7-13700K)
engine\trainer: task=detect, mode=train, model=yolo11s.pt, data=mtga_data/train_data/train_data_50/dataset.yaml, epochs=200, time=Non
Overriding model.yaml nc=80 with nc=4

```

	from	n	params	module	arguments
0	-1	1	928	ultralytics.nn.modules.conv.Conv	[3, 32, 3, 2]
1	-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
2	-1	1	26080	ultralytics.nn.modules.block.C3k2	[64, 128, 1, False, 0.25]
3	-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
4	-1	1	103360	ultralytics.nn.modules.block.C3k2	[128, 256, 1, False, 0.25]
5	-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
6	-1	1	346112	ultralytics.nn.modules.block.C3k2	[256, 256, 1, True]

```

7          -1 1 1180672 ultralytics.nn.modules.conv.Conv [256, 512, 3, 2]
8          -1 1 1380352 ultralytics.nn.modules.block.C3k2 [512, 512, 1, True]
9          -1 1 656896 ultralytics.nn.modules.block.SPPF [512, 512, 5]
10         -1 1 990976 ultralytics.nn.modules.block.C2PSA [512, 512, 1]
11         -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
12        [-1, 6] 1 0 ultralytics.nn.modules.conv.Concat [1]
13         -1 1 443776 ultralytics.nn.modules.block.C3k2 [768, 256, 1, False]
14         -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
15        [-1, 4] 1 0 ultralytics.nn.modules.conv.Concat [1]
16         -1 1 127680 ultralytics.nn.modules.block.C3k2 [512, 128, 1, False]
17         -1 1 147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
18        [-1, 13] 1 0 ultralytics.nn.modules.conv.Concat [1]
19         -1 1 345472 ultralytics.nn.modules.block.C3k2 [384, 256, 1, False]
20         -1 1 590336 ultralytics.nn.modules.conv.Conv [256, 256, 3, 2]
21        [-1, 10] 1 0 ultralytics.nn.modules.conv.Concat [1]
22         -1 1 1511424 ultralytics.nn.modules.block.C3k2 [768, 512, 1, True]
23        [16, 19, 22] 1 820956 ultralytics.nn.modules.head.Detect [4, [128, 256, 512]]
YOLO11s summary: 319 layers, 9,429,340 parameters, 9,429,324 gradients, 21.6 GFLOPs

```

Transferred 493/499 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

train: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_50\labels\train... 45 images, 0 backgrounds, 0 corrupt

val: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_50\labels\val... 5 images, 0 backgrounds, 0 corrupt: 100%

Plotting labels to runs\detect\train4\labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'momentum' au

optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)

Image sizes 640 train, 640 val

Using 0 dataloader workers

Logging results to runs\detect\train4

Starting training for 200 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/200	0G	0.9255	3.168	1.029	172	640: 100% ██████████ 3/3 [00:14<00:00, 4.78s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 1/1 [00:00<00:00, 1.58it/

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/200	0G	0.7013	2.292	0.9305	175	640: 100% ██████████ 3/3 [00:14<00:00, 4.74s/it]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 1/1 [00:00<00:00, 1.59it/

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
3/200	0G	0.5914	1.703	0.8600	173	640: 100% ██████████ 3/3 [00:14<00:00, 4.67s/it]

Sick cards are now being identified at high confidence. Even when they are missed, they are still detected as cards.

Train 7 Screenshot

 Train 7 Screenshot

Looking at the training and validation data we can see that overall performance improves on both overtime. I do not see any concerns with overfitting as the training and validation loss seem to decrease together. Improvement on object classification slowed greatly after a few dozen epochs, but bounding box performance continued to improve steadily. However 200 epochs was sufficient to get enough precision for the application so I stopped it there.

Training & Validation Data

 Training & Validation Data

✖ Training Pass 3 - With Card Backs

```
dataset_path="mtga_data/train_data/train_data_52/dataset.yaml"
```

```
tuned_model = "yolo11s_tuned_52.pt"
```

```
model = YOLO(base_model)
```

```
# Train on annotated dataset
```

```
model.train(
    data=dataset_path,
    epochs=200,
    augment=True,
    mosaic=False,
    hsv_h=0.015,
    hsv_s=0.7,
```

```

hsv_v=0.2,
scale=0.5,
fliplr=0.25
)

# Save the improved model
model.save(tuned_model)

New https://pypi.org/project/ultralytics/8.3.56 available Update with 'pip install -U ultralytics'
Ultralytics 8.3.47 Python-3.11.9 torch-2.5.1+cpu CPU (13th Gen Intel Core(TM) i7-13700K)
engine\trainer: task=detect, mode=train, model=yolo11s.pt, data=mtga_data/train_data/train_data_52/dataset.yaml, epochs=200, time=Non
Overriding model.yaml nc=80 with nc=5

      from  n  params module                                arguments
0         -1  1      928 ultralytics.nn.modules.conv.Conv [3, 32, 3, 2]
1         -1  1     18560 ultralytics.nn.modules.conv.Conv [32, 64, 3, 2]
2         -1  1     26080 ultralytics.nn.modules.block.C3k2 [64, 128, 1, False, 0.25]
3         -1  1     147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
4         -1  1     103360 ultralytics.nn.modules.block.C3k2 [128, 256, 1, False, 0.25]
5         -1  1     590336 ultralytics.nn.modules.conv.Conv [256, 256, 3, 2]
6         -1  1     346112 ultralytics.nn.modules.block.C3k2 [256, 256, 1, True]
7         -1  1     1180672 ultralytics.nn.modules.conv.Conv [256, 512, 3, 2]
8         -1  1     1380352 ultralytics.nn.modules.block.C3k2 [512, 512, 1, True]
9         -1  1     656896 ultralytics.nn.modules.block.SPPF [512, 512, 5]
10        -1  1     990976 ultralytics.nn.modules.block.C2PSA [512, 512, 1]
11        -1  1         0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
12        [-1, 6]  1         0 ultralytics.nn.modules.conv.Concat [1]
13        -1  1     443776 ultralytics.nn.modules.block.C3k2 [768, 256, 1, False]
14        -1  1         0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
15        [-1, 4]  1         0 ultralytics.nn.modules.conv.Concat [1]
16        -1  1     127680 ultralytics.nn.modules.block.C3k2 [512, 128, 1, False]
17        -1  1     147712 ultralytics.nn.modules.conv.Conv [128, 128, 3, 2]
18        [-1, 13]  1         0 ultralytics.nn.modules.conv.Concat [1]
19        -1  1     345472 ultralytics.nn.modules.block.C3k2 [384, 256, 1, False]
20        -1  1     590336 ultralytics.nn.modules.conv.Conv [256, 256, 3, 2]
21        [-1, 10]  1         0 ultralytics.nn.modules.conv.Concat [1]
22        -1  1     1511424 ultralytics.nn.modules.block.C3k2 [768, 512, 1, True]
23        [16, 19, 22]  1     821343 ultralytics.nn.modules.head.Detect [5, [128, 256, 512]]

YOLO11s summary: 319 layers, 9,429,727 parameters, 9,429,711 gradients, 21.6 GFLOPs

Transferred 493/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
train: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_52\labels... 52 images, 0 backgrounds, 0 corrupt: 100%

val: Scanning D:\Dev\MTGA-CV-Voice-Interface\mtga_data\train_data\train_data_52\labels.cache... 52 images, 0 backgrounds, 0 corrupt:
Plotting labels to runs\detect\train5\labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' au
optimizer: AdamW(lr=0.001111, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to runs\detect\train5
Starting training for 200 epochs...

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
1/200   0G       1.007     3.706     1.074     74          640: 100%|██████████| 4/4 [00:17<00:00, 4.30s/it]
          Class  Images  Instances  Box(P)      R      mAP50  mAP50-95): 100%|██████████| 2/2 [00:05<00:00, 2.82s/i

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
2/200   0G       0.7574    2.465     0.9778    56          640: 100%|██████████| 4/4 [00:15<00:00, 3.86s/it]
          Class  Images  Instances  Box(P)      R      mAP50  mAP50-95): 100%|██████████| 2/2 [00:06<00:00, 3.37s/i

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
3/200   0G       0.6466    1.205     0.8947    72          640: 100%|██████████| 4/4 [00:15<00:00, 3.82s/it]

```

Live Labeling

After training the models, I also reviewed their performance by watching their ability to handle real-time inference in a live game. The train_7 model was spotty but the train_50 model was very reliable in identifying cards and buttons.

```

def process_frame(model, frame):
    img = cv2.cvtColor(np.array(frame), cv2.COLOR_BGRA2BGR)

    # Run inference
    results = model.predict(img, conf=0.5, verbose=False)

    # Draw bounding boxes

```

```

for box in results[0].boxes:
    x_min, y_min, x_max, y_max = map(int, box.xyxy[0].tolist())
    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    # Draw rectangle
    cv2.rectangle(img, (x_min, y_min), (x_max, y_max), (0, 255, 0), 2)

    # Draw label
    label = f"{model.names[class_id]} {confidence:.2f}"
    cv2.putText(img, label, (x_min, y_min - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

return img

def live_labeling():
    model = YOLO(tuned_model)

    # Screen capture configuration
    monitor = {"top": 0, "left": 0, "width": 1920, "height": 1080}
    sct = mss()

    cv2.namedWindow("Live Labels", cv2.WINDOW_NORMAL)
    cv2.resizeWindow("Live Labels", 960, 540)

    while True:
        # Capture screen
        screen = sct.grab(monitor)

        # Process frame
        labeled_frame = process_frame(model, screen)

        # Display labeled frame
        cv2.imshow("Live Labels", labeled_frame)

        # Take screenshot
        if cv2.waitKey(1) & 0xFF == ord("s"):
            cv2.imwrite("images/screenshots/screenshot.png", labeled_frame)

        # Exit
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break

    # Cleanup
    cv2.destroyAllWindows()

live_labeling()

```

```

-----
error                                Traceback (most recent call last)
Cell In[4], line 53
     50     # Cleanup
     51     cv2.destroyAllWindows()
--> 53 live_labeling()

```

```

Cell In[4], line 29, in live_labeling()
     26 monitor = {"top": 0, "left": 0, "width": 1920, "height": 1080}
     27 sct = mss()
--> 29 cv2.namedWindow("Live Labels", cv2.WINDOW_NORMAL)
     30 cv2.resizeWindow("Live Labels", 960, 540)
     32 while True:
     33     # Capture screen

```

error: OpenCV(4.10.0) D:\a\opencv-python\opencv-python\opencv\modules\highgui\src\window.cpp:1284: error: (-2:Unspecified error) The function is not implemented. Rebuild the library with Windows, GTK+ 2.x or Cocoa support. If you are on Ubuntu or Debian, install libgtk2.0-dev and pkg-config, then re-run cmake or configure script in function 'cvNamedWindow'

